

5 The pathogen that causes cholera is a prokaryote.

(a) Fig. 5.1 shows an electron micrograph of the pathogen that causes cholera.

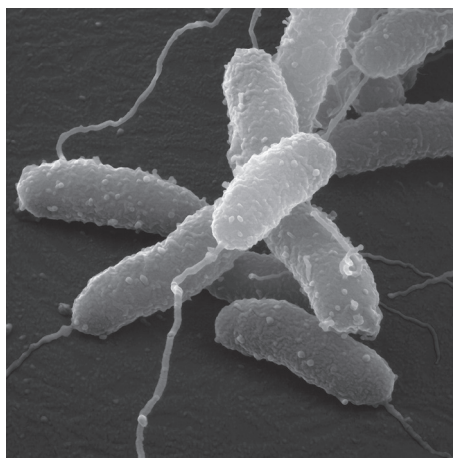


Fig. 5.1

(i) Name the type of electron microscope used to produce the image shown in Fig. 5.1.

..... [1]

(ii) Name the species of prokaryote that causes cholera.

..... [1]

(b) The passage contains a description of the main features of prokaryotic cells. There is **one** factual error in the passage.

Prokaryotic cells are unicellular and generally between $1\mu\text{m}$ and $5\mu\text{m}$ in diameter. Prokaryotes do **not** have organelles surrounded by double membranes. They do have cell surface membranes, 70S ribosomes and a cellulose cell wall. The DNA of a prokaryotic cell is circular and is found free in the cytoplasm rather than enclosed in a nuclear envelope.

Identify **and** correct the factual error in the passage.

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..... [1]

[Total: 3]